Redrob Intelligent Candidate Ranking System

Technical Architecture & Production Verification Deck

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Target Role	Senior AI Engineer — Founding Team (Series A)
Challenge	India Runs Data & AI Challenge — Redrob AI Candidate Ranking
Constraints	CPU-Only | No API Calls | < 5 Min on 100K Profiles | 16GB RAM
Validated On	100,000 real candidate profiles in 19 seconds

Core Design Principles

- **Precision over Recall:** Prefer 10 perfect matches over 1,000 ambiguous leads — explicitly stated in the JD as the evaluation philosophy.
- **Semantic Reasoning:** TF-IDF bigram vectorizer resolves the gap between what the JD *says* and what it *means*, avoiding pure keyword trap matching.
- **Behavioral Realism:** Platform inactivity and low recruiter response rates are first-class scoring signals, not afterthoughts.
- **Anti-Gaming Infrastructure:** Honeypot detection drops fraudulent profiles before tokenization — no compute wasted on fake entries.
- **Production Dominance:** Startup/product company experience is heavily favored; consulting-only and LangChain-only tracks are explicitly penalised.

1. Multi-Dimensional Scoring Framework

Six independently computed components are combined using fixed weights derived directly from the JD priority signals. Each component is capped at 100 before weighting.

Component			
Skill Quality	**28%**	`skill`	Required IR skills (Pinecone, FAISS, NDCG, MRR, Embeddings) filtered by advanced/expert proficiency + endorsements + platform assessment scores.
Experience Trajectory	**25%**	`experience`	5-9 YoE sweet spot. Product company tenure rewarded. Hard penalties for consulting-only, title-chasers (<18mo avg), and non-coding architects.
Semantic (TF-IDF)	**22%**	`semantic`	Bigram TfidfVectorizer (60K features, sublinear_tf=True). Cosine similarity between engineered candidate text and JD text. Advanced skills boosted 3x in token block.
Availability	**15%**	`availability`	Tracks open_to_work_flag, notice_period_days, last_active_date recency, and recruiter_response_rate. Sub-5% response rate is a hard penalty signal.
Platform Trust	**5%**	`platform`	Email/phone verification, profile completeness score, GitHub activity score (0-100) rewarding open-source work.
Location	**5%**	`location`	Pune/Noida = full score. Tier-2 Indian hubs = partial. Relocation-open candidates within India = bonus.

2. TF-IDF Text Engineering Pipeline

To maximize signal quality within CPU-only constraints (no transformer models or GPU), the `build_candidate_text()` function engineers a dense token block before vectorization. Three techniques are applied:

A. Programmatic 3x Term Frequency Boosting (verified in code)

When a required skill (embeddings, Pinecone, FAISS, NDCG, etc.) is matched at **advanced or expert proficiency** level with active deployment tenure, that skill token is appended 3 times into the text block before TF-IDF vectorization. This natively inflates the TF score for high-confidence required skills during cosine similarity computation — without needing a separate embedding model.

Exact implementation from `rank.py`, line 569–570:

```
if is_req and prof in ("advanced", "expert"):
    parts.extend([name, name, name])  # repeat 3x for TF-IDF weight
elif is_req:
    parts.extend([name, name])        # repeat 2x for intermediate
else:
    parts.append(f"{name} {prof}")    # single token for non-required
```

B. Context-Aware Description Truncation

Career descriptions are included at different lengths based on role title relevance. AI/ML/NLP/Search roles get full 500-character descriptions (high signal). Unrelated roles (operations, HR, finance) are truncated to 100 characters — preventing background noise from distorting cosine distance.

```
if any(t in job_title_l for t in ["ml", "ai", "nlp", "data", "search", "engineer"]):
    parts.append(desc[:500])   # full description for AI/ML roles
else:
    parts.append(desc[:100])   # truncated for non-relevant roles
```

C. Vectorizer Configuration

Parameter	Value	Reason
ngram_range	(1, 2)	Captures bigrams like 'vector search', 'hybrid retrieval'
max_features	60,000	Wide vocabulary without memory overflow on 100K profiles
sublinear_tf	True	Dampens effect of repeated generic terms (log scaling)
min_df	1	Keeps rare but important technical terms like 'qdrant', 'weaviate'

3. Adversarial Filter & Disqualification Matrix

The dataset contains ~80 honeypot profiles designed to fool keyword matchers. The `detect_honeypot()` function runs before tokenization — saving compute and preventing fake profiles from polluting TF-IDF vocabulary.

Honeypot Patterns Detected

Pattern		
A — Impossible Breadth	20+ skills at expert level, OR 5+ expert skills with 0 months actual usage	Dropped before scoring
B — YoE Inflation	Self-reported YoE > 3.5x the sum of actual career history durations	Dropped before scoring

Hard Disqualifiers (Immediate Penalty)

- **Unrelated Title Fields:** Marketing Manager, HR Manager, Operations, Accountant, Civil Engineer, Content Writer — immediate -80 score drop.
- **Consulting-Only Career:** Profiles exclusively at TCS, Infosys, Wipro, Accenture, Cognizant, Capgemini, HCL — -30 trajectory penalty.
- **Title Chasers:** Average cross-firm tenure < 18 months — zero stability score.
- **Wrong Domain:** CV/Speech/Robotics/Medical Imaging without any NLP or IR skills — skill score multiplied by 0.3.

Soft Penalties (Score Reduction)

- **Platform Inactivity:** Last active > 180 days — availability score reduced.
- **Low Recruiter Response:** recruiter_response_rate <= 5% — flagged as not actually available despite open_to_work.
- **Long Notice Period:** > 60 days — mild availability penalty.
- **Pure Architecture/Lead:** No production code evidence — experience score capped.

4. Streamlit Interactive Demo (app.py)

The submission includes a fully functional Streamlit web application that wraps the exact `run_ranking()` function from `rank.py` — no separate logic, no duplicate code. Judges can run or view it live at the sandbox link.

Feature		
File Upload / Sample	Upload any .jsonl candidates file or use bundled 300-row sample	st.file_uploader()
Live Progress Log	Streams rank.py step-by-step console output as it runs	progress_cb + st.code()
Top 15 Results Table	Clean ranked table with ID, Title, YoE, Location, Score	st.dataframe()
Overall Score Bar Chart	Plotly bar chart of final scores for top 10 candidates	plotly go.Bar()
Component Breakdown Chart	Grouped bar chart: Skill/Experience/Semantic/Availability per top 5	plotly go.Bar() grouped
Radar Profile Chart	6-dimension radar chart for the #1 ranked candidate	plotly go.Scatterpolar()
Reasoning Expander	Recruiter-style text explanation for each top-5 candidate	build_reasoning() + st.bar_chart()
CSV Download	Full submission.csv (top 100) available for immediate download	st.download_button()

5. Verified Production Results — Top 15 Ranked Candidates

Deterministic output from running `python rank.py --candidates candidates.jsonl --out output/submission.csv` on the full **100,000-candidate dataset in 19 seconds** on a standard CPU machine. Submission validated: 100 rows, ranks 1–100, non-increasing scores, no duplicates.

Rank	Candidate ID	Current Role	Location	YoE	Score	Company
#1	CAND_0018499	**Sr. ML Engineer**	Noida, UP	7	0.9900	Zomato
#2	CAND_0086022	**Sr. Applied Scientist**	Kolkata, WB	5	0.8664	Sarvam AI
#3	CAND_0088025	**Staff ML Engineer**	Jaipur, RJ	9	0.8631	Yellow.ai
#4	CAND_0027691	NLP Engineer	Pune, MH	6	0.7748	Haptik
#5	CAND_0081846	Lead AI Engineer	Jaipur, RJ	7	0.7620	Razorpay
#6	CAND_0046525	Sr. ML Engineer	Pune, MH	6	0.7602	Genpact AI
#7	CAND_0037566	ML Engineer	Bangalore, KA	7	**0.7000**	LinkedIn
#8	CAND_0077337	Staff ML Engineer	Kochi, KL	7	**0.6975**	Paytm
#9	CAND_0039754	Sr. Applied Scientist	Indore, MP	16	**0.6841**	Meta
#10	CAND_0050454	AI Engineer	Delhi, DL	7	**0.6781**	Rephrase.ai
#11	CAND_0064326	Search Engineer	Gurgaon, HR	8	**0.6653**	Sarvam AI
#12	CAND_0039383	Applied ML Engineer	Gurgaon, HR	7	**0.6590**	Meesho
#13	CAND_0055905	Sr. ML Engineer	London	8	**0.6467**	Flipkart
#14	CAND_0044222	AI Engineer	Vizag, AP	8	**0.6462**	PolicyBazaar
#15	CAND_0046064	Sr. NLP Engineer	Coimbatore, TN	9	**0.6439**	Salesforce

* Rank #1 (CAND_0018499) achieves maximum score (0.99) due to: Weaviate + Pinecone + Milvus + Embeddings + Information Retrieval at expert level, 3 AI/ML roles including Senior ML Engineer at Zomato (product company), 15-day notice period, and 95/100 GitHub activity score. Exactly the profile the JD describes.

Submission Validation Status

■ 100 rows in output | ■ Ranks 1–100, no gaps/duplicates | ■ Scores non-increasing | ■ Valid candidate IDs | ■ Correct headers